Teetham Treelayapewat

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Computer Science undergraduate at SUTD with three AI internships spanning computer vision, LLM agents/RAG, and model deployment. Experience with C++/Python, PyTorch, GPU memory optimization, tool-calling systems, and sensor-based engineering.

Education

Singapore University of Technology and Design Sept. 2025 – Present
B.Eng. Computer Science and Design (Full Scholarship) Singapore

Kamnoetvidya Science Academy May 2022 – Mar. 2025
High School Diploma (Full Scholarship) | *AP Computer Science A (Java) & Calculus BC: 5/5* Thailand

Technical Skills

Programming: C++, Python, Java, C, Bash, SQL

AI & Machine Learning: PyTorch, YOLO, OpenCV, Transformers, LoRA, vLLM, RAG, agent tool-calling

Systems & Infrastructure: Linux, Git, Docker, CUDA, SLURM, NVIDIA DGX Spark / H100, LANTA Supercomputer

Hardware & CAD: SOLIDWORKS, Fusion 360, 3D printing, sensor data acquisition

Experience

AltoTech — *AI for building energy management* Aug. 2026 – Oct. 2026
AI Engineer Intern (Onsite, then Remote) Thailand

- Fine-tuned a **30B open model** (NVIDIA Nemotron) with **LoRA** for a **13-tool assistant** designed to run on-premises, finishing with a **two-node run on two DGX Sparks**.
- Curated a **792-example** EN/TH training set from real agent traces; held out rows sharing source documents with test questions.
- Expanded the **agent benchmark** from **179 to 289** EN/TH cases (knowledge, routing, navigation, retrieval, safety), scored by rules plus an LLM judge.
- Improved **hybrid RAG** (vector + keyword) over **41 manuals**: fixed citation pages for text crossing page breaks (**23%** of chunks); top-1 hit rate **55% to 64%**.
- Wrote a **chunked cross-entropy loss** to train on **full 32k-token inputs** on one 80 GB GPU, plus memory guards and a resume fix for two-node training.

BOTNOI Group — *conversational AI company* Jun. 2026 – Aug. 2026
AI Researcher & Development Intern (Remote) Thailand

- Delivered the **store open/close detection** module of a 5-module AI **CCTV monitoring system** for a retail client, combining **YOLO**, **OCR**, and luminance analysis in a rule-based state machine.
- Designed the **proof-of-concept review interface** for detection evidence and set up **Docker Compose** testing across 3 repos.
- Reviewed 5 datasets and 7 papers to scope a **Southeast Asian construction foundation model**, then built door/window and **room segmentation** baselines on 400 floor plans.

Meticuly — *3D-printed medical implants* Jun. 2025 – May 2026
AI Development Engineer Intern (Extended to Remote Role) Thailand

- Developed a **CT classification pipeline** (DICOM → region classification → quality ranking) for the production **CraNeXt** system.
- Reached **99.4%** body-region accuracy (**BiomedCLIP**) and **96.1%** craniectomy accuracy with a fine-tuned **EVA-02**, picked by benchmarking **8 models** (5M–1,012M parameters) on accuracy, speed and size.
- Achieved **1.12 mm error** on **cephalometric X-ray landmark detection** with Thammasat University dentists, using **multi-GPU training** on the **LANTA Supercomputer**.

Selected Projects

High-Energy Particle Simulation (C++ / Geant4) Feb. 2023 – May 2025

- Wrote a **C++** simulation in CERN's **Geant4** to study **Relativistic Runaway Electron Avalanches** under varying magnetic and intermediate fields, tracking particle trajectories and interactions at scale.
- Selected as **Thailand Representative** and Finalist, **Regeneron ISEF 2025**.

Automated Tensile Strength Testing Machine Mar. 2023 – Jun. 2023

- Designed and fabricated an automated physical testing system integrating **embedded control**, **dual-range force sensors**, **real-time data acquisition**, and custom **3D-printed components**.
- Presented at the **18th Siam Physics Congress (SPC2023)**.

Honors, Awards & Leadership

Regeneron ISEF 2025 – Thailand Representative at the world's largest school science fair (Ohio, USA)
Beamline for Schools 2024 – Shortlisted Team, CERN & DESY
ASEAN Data Science Explorers 2024 – Top 10 Finalists, SAP & ASEAN Foundation
SOAR Challenge 2025 – Best Automation Award, SUTD
ThEP Hackathon 2024 – Second Runner-Up, Thailand Center of Excellence in Physics

Vice-President, Eco-Schools Project (WWF, 2024) – led the team to the International Green Flag Award, its highest rating
President, KVIS Science Inspiration Project (2023) – led a 5-day national STEM workshop across 10 schools